



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

WSQ - Generative AI for Problem Solving

TGS Ref No: TGS-2023036653

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 21.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
20.0	1 March 2025	Legacy master lesson plan — Innovative Problem Solving with Generative AI.	Dr. Alfred Ang
21.0	17 August 2026	Major revision. Retitled to the published course title. Rebuilt around 10 real-life workplace case-study activities (ShopFront SG, Meridian Health, Horizon Bank, Nexa Logistics) each with scenario, evidence, discussion questions and debrief. Added the four problem-solving ed-tools (5 Whys, Fishbone, Pareto, System Loop), a 10-prompt GenAI library, and content beefed up from Skills for Success, IMD, MIT CCMIT, Six Sigma, LSE Business Review and current GenAI problem-solving research.	Dr. Alfred Ang

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Course Information

Course Title	WSQ - Generative AI for Problem Solving
WSQ Course Reference	TGS-2023036653
TSC Alignment	Problem Identification (RET-ACE-4006-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, case-study based. Physical classroom, synchronous ZOOM or corporate on-site
Class Size	Maximum 20 learners; activities run in teams of 3-5
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Identify performance deficiencies and root causes impacting organisational aspects
- LO2: Deduce key implications, develop corrective plans, and shortlist viable solutions
- LO3: Determine preferred solutions and evaluate the effectiveness of implemented plans

Target Audience and Prerequisites

- Operations, project, HR, quality, process-improvement and team-leadership roles that must diagnose and resolve recurring workplace problems.
- Beginner level — no prior problem-solving or AI training is required.
- Minimum 3 GCE 'O' Level passes including English, or WPL Level 5.
- At least 1 year of working experience.
- A laptop with a browser and access to a generative AI tool (ChatGPT, Copilot, Gemini or Claude).

Training Resources

- Trainer slide deck (v21.0), Learner Guide and activity briefs on <https://lms-tms.tertiaryinfotech.com>
- Problem-solving ed-tools: 5 Whys (<https://alfredang.github.io/5whys/>); Fishbone (<https://alfredang.github.io/fishbone/>); Pareto Chart (<https://alfredang.github.io/paretochart/>); System Loop (<https://alfredang.github.io/systemloop/>); CollabNote (<https://alfredang.github.io/collabnote/>); Pinboard (<https://alfredang.github.io/pinboard/>)
- Generative AI tools: ChatGPT, Microsoft Copilot, Google Gemini, Claude
- Flip charts, post-it notes and markers for team activities

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), open book.
- Case Study (CS) — an integrated workplace problem-solving scenario, open book.
- Format: Open book covers the slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 2 from 5:45 pm, after the Briefing for Assessment.
- A minimum of 75% attendance based on the SSG Digital Attendance record, and a 'Competent' result in both instruments, are required for certification and funding.

Course Schedule

Day 1 — Frame the problem and find the true root cause — with GenAI as your analyst

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory digital attendance (AM). Slides 1-23
10:00–11:00	60 min	Topic 1 — Problem framing: symptom vs problem vs root cause, well-defined vs ill-defined problems, the IDEAL cycle, the six elements of a workplace problem statement, cognitive barriers, and how GenAI fits the problem-solving cycle. Slides 24-37
11:00–11:15	15 min	Tea break (within training time)
11:15–12:30	75 min	Hands-on: Activity 1: Rewriting a Vague Problem into a Measurable Problem Statement (Slides 38-43)
12:30–13:30	60 min	Lunch break
13:30–14:00	30 min	Digital attendance (PM). Topic 1 continued — the root cause toolkit: choosing between 5 Whys, Fishbone, Pareto and System Loops. Slides 44-49
14:00–15:00	60 min	Hands-on: Activity 2: 5 Whys Root Cause Analysis (Slides 50-54)
15:00–16:00	60 min	Hands-on: Activity 3: Fishbone (Ishikawa) Diagram for Breadth of Causes (Slides 60-64)
16:00–16:15	15 min	Tea break (within training time)
16:15–17:15	60 min	Hands-on: Activity 4: Pareto Analysis to Target the Vital Few (Slides 70-74)

17:15–18:15	60 min	Hands-on: Activity 5: System Loops: Why the Problem Keeps Coming Back (Slides 81-85)
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance. Slide 88

Total training time: 480 minutes (8 hours).

Day 2 — Generate, choose, implement and prove the solution — with GenAI as your consultant

Time	Duration	Topic / Activity
9:30–10:00	30 min	Day 1 recap and mandatory digital attendance (AM). Slides 17-88
10:00–10:45	45 min	Topic 2 — Divergence before convergence: the solution funnel, divergent techniques (brainstorming, SCAMPER, Generic Parts, analogy, nominal group), and separating hard constraints from habits. Slides 89-96
10:45–11:00	15 min	Tea break (within training time)
11:00–12:00	60 min	Hands-on: Activity 6: Divergent Ideation with GenAI (Breadth Before Judgement) (Slides 97-102)
12:00–12:30	30 min	Convergence tools: the Impact-Ease matrix, the weighted decision matrix and sensitivity testing. Slides 103-107
12:30–13:30	60 min	Lunch break
13:30–14:20	50 min	Digital attendance (PM). Hands-on: Activity 7: Converging: Impact-Ease Matrix and Weighted Decision Matrix (Slides 108-112)
14:20–15:10	50 min	Hands-on: Activity 8: Building the Corrective Action Plan (Slides 117-121)
15:10–15:40	30 min	Topic 3 — Implementation and change: factors affecting effectiveness, stakeholder resistance, change management actions and the messenger principle. Slides 123-130
15:40–15:55	15 min	Tea break (within training time)
15:55–16:45	50 min	Hands-on: Activity 9: Implementation Planning and Stakeholder Resistance (Slides 131-135)
16:45–17:30	45 min	Hands-on: Activity 10: Measuring Effectiveness: Did It Actually Work? (Slides 142-146). Sustaining the gain

17:30–17:45	15 min	(PDCA). Slides 147-149 Course summary, Q&A, course feedback and mandatory TRAQOM survey. Slides 150-153
17:45–18:00	15 min	Briefing for Assessment. Slide 154
18:00–18:30	30 min	Digital attendance (Assessment). Final Assessment — Written Assessment (WA, short-answer questions) and Case Study (CS), open book. Slides 155-157

Total training time: 480 minutes (8 hours).

Activity Reference (aligned to TSC abilities)

Topic	TSC coverage	Activities	Case study
Topic 01: Identifying Performance Gaps and Root Causes with Generative AI	K1, K2, K5 · A1, A2	Activity 1, Activity 2, Activity 3, Activity 4, Activity 5	Horizon Bank; Horizon Bank revisited; Meridian Health Screening; Nexa Logistics; ShopFront SG
Topic 02: Developing Corrective Plans and Evaluating Potential Solutions	K4, K6, K9 · A3, A4, A5	Activity 6, Activity 7, Activity 8	Meridian Health
Topic 03: Selecting, Implementing and Measuring Solution Effectiveness	K3, K7, K8 · A6, A7	Activity 9, Activity 10	ShopFront SG

Detailed Activity Schedule

#	Activity	Duration	Grouping	Slides
1	Rewriting a Vague Problem into a Measurable Problem Statement	45 minutes	Teams of 3	Slides 38-43
2	5 Whys Root Cause Analysis	45 minutes	Teams of 3-5	Slides 50-54
3	Fishbone (Ishikawa) Diagram for Breadth of Causes	45 minutes	Teams of 3-5	Slides 60-64
4	Pareto Analysis to Target the Vital Few	40 minutes	Teams of 3-5	Slides 70-74
5	System Loops: Why the Problem Keeps Coming Back	50 minutes	Teams of 3-5	Slides 81-85
6	Divergent Ideation with GenAI (Breadth Before	45 minutes	Teams of 3-5	Slides 97-102

	Judgement)			
7	Converging: Impact-Ease Matrix and Weighted Decision Matrix	50 minutes	Teams of 3-5	Slides 108-112
8	Building the Corrective Action Plan	45 minutes	Teams of 3-5	Slides 117-121
9	Implementation Planning and Stakeholder Resistance	50 minutes	Teams of 3-5	Slides 131-135
10	Measuring Effectiveness: Did It Actually Work?	45 minutes	Teams of 3-5	Slides 142-146